WEEK 1 OF 5 · PLANNING DELIVERABLE

Agribusiness Data Analysis Plan

A structured strategy for sourcing, cleaning and analysing publicly available agricultural data to forecast market trends, with a five-week execution timeline and defined quality gates.

PREPARED BY	Data Science Intern
PROGRAMME	Junior Data Scientist Internship, Nagpur
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SCOPE	HORIZON	EVIDENCE BASE
6 crops · 8 states	**12 months**	**38 public sources**
India-first, globally benchmarked	Nov 2026 – Oct 2027	6 data families

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Document control

Table 1. *Document identification and control record.*

Item	Detail
Title	Agribusiness Data Analysis Plan
Deliverable	Week 1 task — planning document
Version / status	1.1 — revised; Appendix F added
Author	Data Science Intern, Junior Data Scientist Internship, Nagpur
Date of issue	27 October 2026
Review cycle	Revised at each weekly milestone (M1–M5)
Scope of analysis	India primary; FAOSTAT, USDA and OECD-FAO used as global benchmarks
Companion artefacts	Source register (D2), data dictionary (D4), notebooks (D5–D8)
Distribution	Internship evaluator; internal project repository

HOW TO READ THIS PLAN

Sections 2–5 cover the reasoning: why it matters, which data exists, which indicators carry signal and how they are modelled. Sections 6–8 are the operating plan — sequence, dated timeline and artefacts.

1. Executive Summary

This document is the Week 1 deliverable of a five-week internship and the planning spine for the four that follow. It commits to a defined question, a defined evidence base and a defined sequence of work, rather than describing agribusiness analytics in general terms. The output it specifies is a forecast-ready panel of public data, a set of indicators that carry economic signal, a benchmarked forecasting model and a decision memo an operations manager could act on.

Three constraints shaped the design. Everything must come from public sources, so no proprietary price feed or paid satellite subscription can sit on the critical path. The work must fit inside five part-time weeks, so the plan is deliberately narrow: six crops, eight states, one primary forecast target. And the later weekly tasks are not independent essays but stages of the same pipeline, so each phase here produces an artefact that the next task consumes directly.

> **SCOPE DECISION**
> The analysis covers six crops across eight Indian states, benchmarked against global series rather than pooled with them. The six crops were chosen to represent six distinct commercial behaviours — a staple cereal (wheat), a water-intensive cereal (paddy), a feed grain (maize), an oilseed (soybean), a fibre and export crop (cotton) and a perishable horticulture crop (onion). Pooling a US corn-belt series with an Indian mandi series produces a model that fits neither; the global data is therefore used to sanity-check levels and trends, not as training rows.

1.1 The plan at a glance

The table below is the one-page contract. Everything later in the document elaborates one of these rows.

Table 2. *The plan in nine decisions.*

Dimension	Decision	Why this choice
Geography	India, with eight states: Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Maharashtra, Karnataka, Andhra Pradesh and West Bengal	Covers the main surplus belts plus two large consuming states; all eight have usable mandi and yield series
Crops	Wheat, paddy, maize, soybean, cotton and onion	One crop per commercial behaviour: staple, water-intensive staple, feed, oilseed, fibre/export, perishable horticulture
Time span	FY 2015-16 to FY 2025-26 annual; January 2019 to October 2026 daily prices	Long enough to contain three regime shifts: COVID-19, the 2021-22 input-cost spike and the 2023 export restrictions
Frequency	Monthly for prices; annual for production; daily retained for volatility only	Matches the cadence at which each decision is actually taken
Forecast target	Monthly modal mandi price per crop, 1–12 months ahead; annual yield per crop and state	Price and yield are the two variables every downstream task depends on
Horizon	Reported separately at 1, 3, 6 and 12 months	Short horizons are genuinely forecastable; twelve months is a scenario range, not a point promise
Unit of analysis	State × crop × month for prices; state × crop × year for yield	Supports time-series and cross-sectional comparison with the same panel
Benchmark	Two naive forecasts (random walk and seasonal naive), then exponential smoothing, then exogenous models	A model must beat both naive forecasts, not just the seasonal one — the pilot in Appendix F shows why
Tooling	Python (pandas, statsmodels, LightGBM), DuckDB, Git, Streamlit	Free, reproducible and sufficient for the data volume involved

1.2 What the plan is designed to produce

- **A single analysis-ready dataset —** one documented, versioned panel that any analyst can rebuild from the raw snapshots in under ten minutes;
- **A defined indicator set —** 30 indicators across five families, each with a written formula, source, frequency and failure mode;
- **A benchmarked forecast —** a forecast compared honestly against the naive benchmarks, with intervals and a stated reason for every crop where it fails;
- **A decision memo —** two pages stating what the numbers imply, for whom, and what would change the conclusion.

1.3 How success will be judged

The project succeeds if five tests pass. They are deliberately about reliability rather than sophistication, because an evaluator or a manager can verify each one.

- **Coverage —** at least 90% of the planned series ingested, each with retrieval date, licence and endpoint recorded;
- **Integrity —** at least 95% completeness on the core panel after documented cleaning rules, with every remaining gap classified rather than silently imputed;
- **Skill —** MASE below 1.0 against the better of the two naive benchmarks on at least four of six crops at the three-month horizon, and reported honestly where it is not;
- **Traceability —** every chart in the final report reproducible from a named query on a named data version;
- **Legibility —** a reader with no agronomy background can follow the memo without a glossary.

1.4 Evidence already in hand

A plan that promises a method is weaker than a plan that has already run it once. Before any Indian series was touched, the pipeline described in Sections 5 and 6 was run end to end on the World Bank's monthly commodity price file — a public series that was downloadable on the day this document was written. Appendix F reports the work in full and states its limits; three results matter here.

- **The arithmetic works on real data —** On 93 monthly observations across five crops, a seasonal index, a storage-carry calculation and a rolling-origin backtest all produced numbers a manager could act on. The method risk in Weeks 3 and 4 is therefore mostly removed before Week 2 begins.
- **Seasonality is worth money where it is real —** Wheat's December average sits 17.1% above its August average, or US$40/t on a US$237/t trough. That is the size of the choice a storage or procurement desk actually faces, and the reason this plan reports a spread and a range rather than a direction.
- **The original benchmark was too generous —** Against seasonal naive the seasonal model won on 90.8% of origins at one month. Against a random walk it lost in 45 of 60 crop-horizon cells. The benchmark in Section 5.5 is now a pair, and the promotion rule requires beating both.

2. Research Rationale

2.1 Why agricultural forecasting is worth the effort

Agricultural supply is committed months before demand is known. Area is sown on an expectation, weather then decides the yield, and policy decides whether the crop may leave the district. This sequence — expectation, then biology, then administration — is what separates agribusiness from most other markets, and it is why a forecast has value even when it is imprecise: it narrows the range of outcomes before money is committed.

In India the sequence is unusually visible. A fragmented supply base sells through a large, unevenly digitised network of regulated wholesale markets; a minimum support price puts a floor under some crops and distorts the planting decision for others; and export policy can be changed by notification. Onion is the standard illustration — a perishable crop with no meaningful storage buffer, where a few weeks of supply imbalance can move prices by a multiple rather than a percentage.

The global series show the same behaviour in slower motion, and the magnitudes are worth stating. US wheat traded at US$197/t in April 2019 and US$447/t in March 2022; cotton ran from US$1.40/kg in April 2020 to US$3.60/kg in May 2022; Thai rice fell from US$660/t in January 2024 to US$356/t in October 2025, a 46% decline over twenty-one months, before recovering to US$482/t. Appendix F computes these figures from the raw file. None of the three moves was signalled by the price a year earlier — which is the case for doing this work at all, and the case for measuring it against a naive forecast rather than presenting a curve without one.

The commercial stake is thin margins. Agri-input retail, procurement, warehousing and crop lending typically earn single-digit percentage margins on turnover, so a ten per cent error in an assumed price can erase the entire margin on a transaction. That arithmetic is the reason this plan spends its effort on indicator quality and data provenance rather than on model complexity: a well-understood seasonal pattern with a confidence interval is worth more than an opaque model with a decimal place.

2.2 Who acts on the output

A forecast without a named decision is a curiosity. The table below fixes the audience before any data is collected, which in turn fixes the required granularity and refresh rate.

Table 3. *Decision-makers, the decisions they face, and the indicators that serve them.*

Decision-maker	Decision it informs	Indicators that matter	Cadence
Procurement / trading desk	When and where to buy, and how much inventory to carry	Modal price versus three-year seasonal, arrivals, stock-to-use	Weekly
Farmer or farmer producer organisation	Sell at harvest or store, and at which market	Price seasonality, storage cost, MSP floor, local basis	At harvest
Agri-input retailer or distributor	Seed and fertiliser stocking before the season	Sown area, crop-price-to-input-price ratio, credit availability	Monthly, pre-season
Lender or crop insurer	Whether to lend, at what limit, and how to price risk	Yield trend, rainfall departure, vegetation anomaly, claim history	Seasonal
Processor or exporter	Contract timing and export parity	Futures basis, export policy status, exchange rate, freight	Weekly
Buffer agency or policy analyst	Procurement volume and release timing	Production estimate, public stocks, pass-through to retail inflation	Monthly

2.3 The forecastability principle

Agricultural series do not share one statistical character, and treating them as though they do is the most common design error in this field. A monthly onion price and an annual wheat yield are not variations of the same problem. The plan therefore separates three tiers and commits to a different claim for each.

Table 4. *Three forecastability tiers and the claim permitted for each.*

Tier	Series in this tier	Appropriate method	What is claimed
A — Forecastable	Monthly wholesale prices, arrivals, spatial price spreads	Seasonal decomposition, SARIMAX with exogenous drivers, gradient boosting; prediction intervals	A point forecast with an 80% and 95% interval, 1–6 months ahead

Tier	Series in this tier	Appropriate method	What is claimed
B — Partly forecastable	Annual area, production and yield per state	Trend plus weather regression, distributed lags; banded output	A range and a direction, stated before harvest and revised at each official estimate
C — Not forecastable	Export bans, duty changes, MSP revisions, disease outbreaks	Event study and scenario analysis; no statistical point forecast	Conditional statements: if policy X occurs, the effect is approximately Y

The discipline this imposes is the point. Tier A results are published as forecasts. Tier B results are published as ranges with an explicit confidence statement. Tier C results are published as scenarios. Collapsing Tier C into Tier A is precisely how confident and wrong forecasts are produced, and it is the failure the evaluation protocol in Section 5.5 is built to catch.

2.4 Research questions and hypotheses

Five questions organise the analysis. Each has a test attached, so that the later tasks inherit a measurable objective rather than a topic.

Table 5. *Research questions and the test attached to each.*

ID	Research question	How it will be tested
RQ1	Which indicators lead wholesale price movements by one to six months, and how stable is that lead across crops?	Cross-correlation on first differences, estimated on rolling five-year windows to test stability
RQ2	How much of district-level yield variation is explained by agro-climatic variables compared with input intensity?	Nested regressions with variance decomposition; collinearity checked before interpretation
RQ3	Can a parsimonious model beat both naive benchmarks at three and six months?	MASE over rolling forecast origins, reported per crop and per horizon
RQ4	Is price transmission from wholesale to retail symmetric, or do margins widen faster than they narrow?	Asymmetric error-correction model on log price pairs
RQ5	Which shocks — rainfall, fertiliser cost, export policy — move gross margin most?	Elasticity-based scenario grid with a stated parameter source for each elasticity

Four working hypotheses follow from these questions. Each is falsifiable and each is expected to fail for at least one crop — which is itself a finding worth reporting.

- **H1 —** a rainfall deficit during the reproductive window lowers yield more than an equal deficit at sowing. Tested through distributed-lag coefficients on fortnightly rainfall.
- **H2 —** spatial price dispersion widens when arrivals fall below their twentieth percentile, indicating thin-market behaviour rather than transport cost.
- **H3 —** fertiliser price spikes pass into acreage with a one-season lag, not within the same season.
- **H4 —** onion prices are forecastable one month ahead but not twelve, while wheat prices show the opposite pattern because of buffer-stock policy.

2.5 Where agricultural forecasts usually fail

The failure modes below are drawn from the structure of the data itself, not from generic modelling advice. Naming them in advance is what makes the cleaning rules in Section 6.4 and the evaluation protocol in Section 5.5 auditable.

Table 6. *Eight failure modes, their symptoms, and the control that addresses each.*

Failure mode	How it appears in practice	Mitigation built into this plan
Look-ahead through revised data	The model appears to predict a yield that was revised upward two years after the harvest it describes	Every official release stored as a vintage; models trained only on the vintage available at the forecast origin

Failure mode	How it appears in practice	Mitigation built into this plan
Unit and definition drift	Quintals mixed with tonnes, acres with hectares, nominal with real, fiscal year with calendar year	One canonical unit and calendar table, applied at ingestion; nominal and real reported side by side
Aggregation bias	A national trend that contradicts the trend in every individual state	Models estimated at the level where the decision is made; state weights checked before any national claim
Coverage bias	Mandi prices that reflect regulated markets only, while a growing share of trade moves outside them	Population stated explicitly; arrivals used as weights; cross-checked against eNAM and retail series
Spurious seasonality	A three-year series that confirms a pattern which is really one bad monsoon	Minimum of eight years required before a seasonal pattern is claimed; stability tested across sub-periods
Regime shift	An export ban that breaks the price relationship overnight and never restores it	Structural-break tests; scenario mode replaces point forecasting after a break
Thin markets	Zero arrivals, a single quote for the day, or a price typed as 400 instead of 4,000	Minimum-observation rule per market; robust filters that flag rather than delete
Leakage in feature engineering	A scaler or rolling mean computed over the full history, so the past sees the future	All transforms fitted inside each training fold; a leakage check in the pipeline test

3. Data Landscape and Source Inventory

Public agricultural data is abundant, disaggregated and inconsistent. The same quantity — wheat production in Madhya Pradesh — exists in at least four places, at four levels of aggregation, with four different publication lags and occasionally four different values. The inventory below is therefore organised by what each source can carry into the model, not by which agency publishes it. Thirty-eight sources across six families are catalogued; twelve sit on the critical path.

3.1 Selection criteria

Every source was screened against eight criteria before being admitted to the register. The criteria are applied in order, and a source that fails the first three is recorded but not used.

- **Public and free at the point of use —** no paywall, no institutional subscription, no request-by-email process on the critical path.
- **Documented provenance —** the publishing agency, collection method and revision policy are identifiable.
- **Sufficient granularity —** the unit of observation is at or below the level at which the decision is taken.
- **Adequate cadence —** the refresh rate is at least as fast as the decision cycle it informs.
- **Machine readability —** CSV, JSON or a documented API, rather than a PDF table that must be re-keyed.
- **Continuity —** at least eight years of comparable history, with definitions held constant.
- **Permissible use —** terms permit reuse with attribution; no personal data; no access behind a login that would breach terms.
- **Redundancy —** every critical series is carried by at least two independent publishers, so that a single agency's revision cannot silently move the result.

> **THE REDUNDANCY RULE**
>
> Prices are covered by Agmarknet, the data.gov.in API and eNAM; production by the Directorate of Economics and Statistics and FAOSTAT; global benchmarks by USDA and OECD-FAO. Where two sources disagree by more than 5%, the discrepancy is logged as a finding rather than averaged away — the gap between them is often the most informative number in the dataset.

3.2 Family 1 — Production, area and yield

Table 7. *Production, area and yield sources. DES is the primary series; FAOSTAT and USDA provide the cross-check.*

Source	What it provides	Granularity · refresh	Access	Role
USDA NASS Quick Stats	Planted and harvested area, production and yield for US crops, with a published survey methodology	State and county · annual, plus monthly Crop Progress	Free API key	Benchmark
USDA WASDE and PSD	Monthly world supply-and-demand balances, ending stocks and stock-to-use ratios	Country and commodity · monthly	Free download	Benchmark
FAO FAOSTAT	Area harvested, production, yield, producer prices and food balance sheets for about 245 countries	Country · annual	Free API and bulk CSV	Benchmark
Directorate of Economics and Statistics (DES)	State- and district-level area, production and yield, including advance estimates released in successive rounds	State and district · annual, multi-vintage	Free download	Core supply
Agriculture Census	Holding size, tenancy, irrigation and cropping structure	District · five-yearly	Free download	Context
OECD-FAO Agricultural Outlook	Ten-year projections of production, consumption, trade and prices	Country and commodity · annual	Free summary; paid full data	Prior
Eurostat and DG AGRI	EU production, prices and short-term outlook, useful as a second temperate-crop benchmark	Member state · monthly and annual	Free API	Optional

3.3 Family 2 — Prices and market arrivals

Table 8. *Price sources. Agmarknet supplies the forecast target; futures and global prices supply leading information.*

Source	What it provides	Granularity · refresh	Access	Role
Agmarknet	Daily quotes from regulated wholesale markets: minimum, maximum and modal price, with arrivals by commodity and variety	Mandi · daily	Free portal, no API	Core price
data.gov.in (Agmarknet resource)	The same mandi data exposed as a JSON API with resource identifiers and filters	Mandi · daily	Free API key, rate-limited	Ingestion
eNAM	Trade volumes and prices from mandis onboarded to the electronic national market	Mandi · daily	Free portal	Validation
World Bank Pink Sheet	Global commodity prices, nominal and real, for grains, oils, fibres and fertiliser inputs	Commodity · monthly	Free Excel	Benchmark
FAO Food Price Index	Index of five food commodity groups, with a deflated variant	Global · monthly	Free	Macro context
NCDEX and MCX	Domestic agricultural futures settlement prices, open interest and contract curves	Contract · daily	Free delayed quotes	Forward signal
CME Group	Global grain, oilseed and fibre futures, used for export-parity calculations	Contract · daily	Free delayed quotes	Parity
AGMARK and Department of Consumer Affairs	Retail and wholesale prices for essential commodities in major centres	Centre · daily to weekly	Free portal	Margin analysis

3.4 Family 3 — Inputs and resource utilisation

Table 9. *Input and resource-use sources. These carry the cost side of the margin calculation.*

Source	What it provides	Granularity · refresh	Access	Role
Department of Fertilizers / FAI	Monthly nitrogen, phosphorus and potassium consumption by state and product	State · monthly	Free reports	Cost driver
CACP cost of cultivation	A2+FL and C2 costs per hectare and per quintal, by crop and state	State · annual	Free reports	Margin base
Central Ground Water Board	Water-level trends and stage of groundwater extraction by block	Block · annual	Free reports	Resource risk
India-WRIS (Ministry of Jal Shakti)	Reservoir storage and irrigation command-area data	Reservoir · weekly	Free portal	Water supply
PMFBY	Crop insurance enrolment, claims and weather-trigger outcomes	State, crop and season · annual	Free dashboards	Risk proxy
PPAC and IEA	Diesel, electricity and fertiliser feedstock prices	National · monthly	Free and paid tiers	Cost shock

3.5 Family 4 — Agro-climate and earth observation

Table 10. *Agro-climate sources. Weather and vegetation indices are the leading indicators for the yield model.*

Source	What it provides	Granularity · refresh	Access	Role
India Meteorological Department	Daily rainfall and temperature, cumulative monsoon departure by subdivision and district	District and subdivision · daily	Free portal	Yield driver
IMD Agromet Advisory Services	Block-level advisories, pest and disease warnings by crop	Block · twice weekly	Free	Context
NASA POWER	Daily temperature, precipitation, humidity and solar radiation on a 0.5° grid, with a long archive	Grid cell · daily	Free API, no key	Gap filling
Copernicus Sentinel-2	10 m optical imagery for NDVI, NDWI and crop-condition monitoring	10 m · about 5-day revisit	Free, registration	Nowcast
MODIS and VIIRS	250 m vegetation indices as 16-day composites with a twenty-year history	250 m · 16-day	Free	Long baseline
NBSS&LUP	Soil texture, depth and taxonomic class	Polygon · static	Free maps	Yield potential
NOAA Oceanic Niño Index	El Niño–Southern Oscillation phase and anomaly	Global · monthly	Free	Seasonal outlook

3.6 Family 5 — Trade, stocks and policy

Table 11. *Trade and policy sources. Policy events are recorded as dated flags, not as continuous variables.*

Source	What it provides	Granularity · refresh	Access	Role
UN Comtrade	Import and export values and quantities by HS code and partner country	Reporter–partner · monthly and annual	Free tier with quota	Trade flow
ITC Trade Map	Trade competitiveness indicators by product and destination market	Country and product · annual	Free registration	Market access
APEDA	India's agricultural export data by commodity and destination	Commodity · monthly	Free	Export series
DGFT	Export bans, duty changes and trade notifications, with effective dates	Event-based	Free	Regime flag

Source	What it provides	Granularity · refresh	Access	Role
CACP minimum support prices	Announced MSP by crop and marketing year, with bonus and procurement status	National · annual	Free	Policy floor
FEWS NET and FAO GIEWS	Food-security alerts and crop-prospect assessments	Region · monthly	Free	Shock context

3.7 Family 6 — Survey, credit and structural data

Table 12. *Structural and contextual sources. These inform priors and interpretation rather than model features.*

Source	What it provides	Granularity · refresh	Access	Role
NSSO Situation Assessment Survey	Household income, consumption, indebtedness and asset ownership among agricultural households	State · roughly five-yearly	Free microdata	Demand structure
RBI and NABARD	Agricultural credit flow, rural wages and agricultural gross value added	National and state · quarterly	Free	Macro context
ICAR and IFPRI research	Yield gaps, technology adoption and agronomic trial results	Study level · irregular	Free	Priors
PIB and business press	Policy announcements and market reporting for event extraction	Article · daily	Free	Event flags

3.8 Access verification

Every URL was checked on 27 October 2026 before being written into the source register. The table below records the outcome. Automated checks returned HTTP 403 for two commercial sites that load normally in a browser; they are retained with that caveat rather than excluded. Certain ministry subdomains are not reachable from the audit environment used for this check and will be confirmed from an Indian network on day three of Week 2.

Table 13. *Access check performed 27 October 2026. A 403 indicates bot protection, not an unavailable source.*

Source	Domain	Response
NASS Quick Stats API	quickstats.nass.usda.gov/api	200
FAOSTAT	fao.org/faostat/en/	200
FAO Food Price Index	fao.org/worldfoodsituation/foodpricesindex/en/	200
World Bank agriculture data	data.worldbank.org/topic/agriculture-and-rural-development	200
World Bank Commodity Markets	worldbank.org/en/research/commodity-markets	200
Agmarknet	agmarknet.gov.in	200
data.gov.in	data.gov.in	200
eNAM	enam.gov.in/web/	200
DES, Ministry of Agriculture	desagri.gov.in	200
IMD Mausam	mausam.imd.gov.in	200
IMD Agromet Advisory	agromet.imd.gov.in	200
NASA POWER	power.larc.nasa.gov	200
Copernicus Data Space	dataspace.copernicus.eu	200
UN Comtrade	comtradeplus.un.org	200
ITC Trade Map	trademap.org	200
Eurostat agriculture	ec.europa.eu/eurostat/web/agriculture	200
NCDEX	ncdex.com	200
Central Ground Water Board	cgwb.gov.in	200

Source	Domain	Response
PMFBY	pmfby.gov.in	200
APEDA	apeda.gov.in	200
CME Group	cmegroup.com/markets/agricultural.html	403 to scripts
OECD-FAO Outlook	oecd.org/en/publications	403 to scripts

3.9 Known gaps and the proxy chosen for each

No public dataset answers every question. Rather than pretend otherwise, each gap is named with the substitute that will be used and the bias that substitute introduces.

Table 14. *Six data gaps and the substitute adopted for each.*

Gap	Consequence if ignored	Proxy and stated limitation
No free real-time national stock figures	A supply shock is mistaken for a demand shift	Stock-to-use implied from USDA PSD balances and the production–trade identity; treated as an estimate with a wide band
District yield published with a one- to two-year lag	The most recent year is missing exactly when it matters	Vegetation and weather nowcast, later reconciled against the advance estimates; nowcast error quantified at each vintage
Mandi coverage favours regulated markets	Price levels and trends may not represent total trade	Population defined as regulated mandis, with arrivals as weights; validated against eNAM and retail series
No farm-gate price series	The farmer's realised price is not directly observable	Modal mandi price used as a proxy, with the commission and transport wedge documented rather than assumed away
Arrivals reported in mixed units	Volume comparisons across states are invalid	Commodity-specific bag and lot weights normalised to tonnes, with a per-commodity conversion table held under version control
Policy shocks are not continuous variables	A regression attributes a policy break to weather	Dated event flags plus scenario analysis; no attempt to fit a coefficient to a single policy episode

3.10 Legal, ethical and licensing position

- **Terms of use —** every source is used within its published terms; automated requests are throttled and identified, and no portal is scraped behind a login.
- **Attribution —** the originating agency is credited in every chart and table, with the retrieval date attached to the underlying snapshot.
- **Personal data —** the dataset contains no individual or household identifiers; survey microdata, if used at all, is aggregated before storage.
- **Honest limitation —** the final report states what the data cannot show, including the coverage limitations above, rather than leaving them to be discovered.
- **Reproducibility —** every transformation applied to a series is recorded in the repository, so that any published figure can be retraced from the raw snapshot.

4. Indicator Framework

An indicator earns its place only if it changes a decision. The framework below sorts candidate metrics into five lenses and then classifies each one as a forecast target, a driver, a risk flag or context. This prevents the common failure of a dashboard with forty charts and no argument.

4.1 Five lenses

Table 15. *Five analytical lenses and the role each plays in the forecast.*

Lens	Question it answers	Representative indicators	Role in the model
Supply	How much will be produced, and when?	Area sown, production, yield, sowing progress, vegetation anomaly	Target
Price	What is the market paying, and how is that changing?	Modal price, real price, arrivals, dispersion, MSP premium, volatility	Target
Demand and trade	Where will the crop go, and what is already committed?	Per-capita availability, stock-to-use, net trade, policy status	Driver
Resource and cost	What does it cost to produce, and is the resource base sustainable?	Fertiliser intensity, irrigation intensity, groundwater stress, cost per quintal	Driver
Risk and context	What could break the forecast?	Rainfall departure, heat-stress days, dry spells, ENSO phase, pest alerts, policy events	Risk flag

4.2 Supply indicators

Table 16. *Supply-side indicators.*

Indicator	Definition or formula	Freq.	Primary source	Role
Area sown	Reported area brought under the crop, by state and season	Weekly in season	DES / state bulletins	Driver
Production	Official estimate of output in tonnes, retained by vintage	Annual	DES	Target
Yield	Production (t) ÷ area harvested (ha)	Annual	Derived	Target
Yield deviation	(Actual − five-year trend) ÷ trend × 100	Annual	Derived	Target
Sowing progress	Reported % area sown − normal % for the same week, in percentage points	Weekly	Ministry weekly note	Driver
Vegetation anomaly	(NDVI − five-year same-dekad mean) ÷ standard deviation	10-day	Sentinel-2, MODIS	Driver

4.3 Price indicators

Table 17. *Price-side indicators, including the two forecast targets.*

Indicator	Definition or formula	Freq.	Primary source	Role
Modal price	Most frequently traded price at the mandi, ₹ per quintal	Daily → monthly	Agmarknet	Target
Real price	Nominal price ÷ CPI (base 2011-12) × 100	Monthly	Derived from CPI	Target
Seasonal index	Month mean ÷ twelve-month mean × 100	Monthly	Derived	Descriptive
Arrivals	Quantity arriving at the mandi, normalised to tonnes	Daily → monthly	Agmarknet	Driver
Spatial dispersion	(P90 − P10) ÷ P50 across mandis for the same crop and date	Weekly	Derived	Driver
MSP premium	(Modal price − MSP) ÷ MSP × 100	Monthly	CACP and Agmarknet	Context
Volatility	Standard deviation of log monthly change × √12	Monthly	Derived	Risk

4.4 Demand and trade indicators

Table 18. *Demand, trade and policy indicators.*

Indicator	Definition or formula	Freq.	Primary source	Role
Per-capita availability	(Production + imports − exports − feed and waste) ÷ population	Annual	FAOSTAT balances	Driver
Stock-to-use ratio	Ending stocks ÷ total domestic use × 100	Monthly	USDA PSD	Driver
Net trade	Exports − imports, by HS code, in tonnes and value	Monthly	Comtrade, APEDA	Driver
Retail–wholesale spread	(Retail − wholesale) ÷ wholesale × 100	Weekly	AGMARK / DoCA	Context
Policy status flag	Enumerated state: open, restricted or prohibited, with effective date	Event	DGFT notifications	Risk

4.5 Resource utilisation and cost indicators

Table 19. *Resource-use and cost indicators, which carry the cost side of gross margin.*

Indicator	Definition or formula	Freq.	Primary source	Role
Fertiliser intensity	N + P + K nutrient consumption (kg) ÷ gross cropped area (ha)	Monthly → annual	Department of Fertilizers	Driver
Nutrient balance	Actual N:P:K ratio compared with the recommended ratio for the crop	Annual	Derived	Context
Irrigation intensity	Gross irrigated area ÷ net sown area × 100	Annual	DES / Ag Census	Driver
Groundwater stress	Stage of groundwater extraction, by block (%)	Annual	CGWB	Risk
Cost per quintal	Cost of cultivation (A2+FL) ÷ yield in quintals per hectare	Annual	CACP	Context
Farm power	Installed kW ÷ net sown area (kW per ha)	Annual	DES	Context

4.6 Risk and context indicators

Table 20. *Risk and context indicators, used as model flags rather than continuous regressors where sparse.*

Indicator	Definition or formula	Freq.	Primary source	Role
Rainfall departure	(Actual − normal) ÷ normal × 100, cumulative over the crop window	Daily → weekly	IMD	Driver
Heat-stress days	Days with maximum temperature above the crop's threshold during grain filling	Daily	IMD, NASA POWER	Driver
Dry-spell length	Longest run of days with rainfall below 2.5 mm inside the crop window	Daily	IMD, NASA POWER	Risk
ENSO phase	Oceanic Niño Index value and categorical phase	Monthly	NOAA	Context
Pest and disease alerts	Count of active advisory flags for the crop and district	Twice weekly	IMD Agromet	Risk
Crop-calendar shift	Deviation of the harvest date from the five-year median, in days	Season	Derived	Context

4.7 Prioritisation: what gets built first

Thirty indicators is more than five weeks can carry to production quality. The prioritisation below is a build order, not a wish list: the must-have set alone supports a defensible forecast, and the rest are added only if a milestone gate is met early.

Table 21. *Build order for the thirty indicators.*

Priority	Indicators	Rationale
Must have (12)	Modal price, arrivals, real price, production, yield, area sown, rainfall departure, vegetation anomaly, MSP premium, stock-to-use, fertiliser intensity, cost per quintal	These twelve support the price and yield forecasts and the margin calculation; without them no result is reportable
Should have (10)	Seasonal index, volatility, spatial dispersion, retail–wholesale spread, irrigation intensity, groundwater stress, net trade, per-capita availability, heat-stress days, sowing progress	These explain why a forecast moved and give the memo its argument
Nice to have (8)	Nutrient balance, farm power, ENSO phase, pest alerts, policy status flag, dry-spell length, crop-calendar shift, yield deviation	Useful for narrative and for the following internship tasks, but not load-bearing for the Week 1–5 outputs

4.8 Data quality measures attached to the indicators

Each indicator carries a quality record. A metric that is 80% complete but 100% timely is more useful than one that is complete and six months late, and the difference must be visible rather than buried in a footnote.

Table 22. *Six quality dimensions, their thresholds and the response when a threshold fails.*

Dimension	Definition	Threshold	Action if breached
Completeness	Share of expected observations present in the core panel	≥ 95%	Investigate the gap, document cause, apply the rule for that crop
Timeliness	Days between source publication and availability in the panel	≤ lag + 2 days	Automate the pull; flag late series in the dashboard
Validity	Share of price observations inside a crop-specific plausible band	100%	Hold the outlier for review; never auto-correct silently
Uniqueness	Duplicate records after the deduplication rule	0	Fail the pipeline run; fix the key, not the symptom
Consistency	Unit and calendar mismatches caught by the validation suite	0	Block publication until resolved
Stability	Absolute revision between successive official vintages	< 5%	Record the revision; widen the interval where revisions are large

5. Methodology and Analysis Design

The analytical design follows one rule: start with the simplest thing that could work, and only add complexity when a backtest proves it earns its place. Naive forecasts are harder to beat than they look, and which one is hardest depends on the crop and the horizon: the pilot in Appendix F found a seasonal model that beat the seasonal naive benchmark on nine origins in ten at one month and still lost to last month's price in most cells. Most of the modelling effort therefore goes into choosing the right benchmark and beating it honestly, rather than into choosing an algorithm.

5.1 Unit of analysis and panel structure

Two linked tables carry the analysis. The annual table is keyed on state, crop and agricultural year and holds area, production and yield. The monthly table is keyed on state, crop and month and holds prices, arrivals and weather aggregates. They join on state and crop, with the annual value allocated to the months of the crop's marketing window rather than to the calendar year.

- **One storage shape —** long (tidy) format for storage; wide format only inside a model script, where it is reproducible from the long table.

- **Provenance is a column —** every fact carries source_id, vintage_id and retrieved_at. A number without those three fields is not admitted to the panel.
- **Vintages are rows —** production estimates are stored by release, not overwritten, so the panel can reconstruct what was known on any given date.
- **Geographies are coded —** LGD codes for states and districts, a crop master table, and a mandi register; names are labels, codes are keys.
- **Time is versioned —** the panel is not a single file but a set of dated snapshots, so a result can always be reproduced against the data that produced it.

5.2 The analytical sequence

Seven steps, in order. Each produces something the next step can use, and none is skipped when results look disappointing.

- **Describe —** distributions, coverage by state and crop, and the shape of every gap. The purpose is to find what is absent before modelling what is present.
- **Decompose —** separate trend, season and residual with STL; check whether the seasonal pattern is stable across sub-periods.
- **Test relationships —** stationarity tests, lead–lag cross-correlations, elasticity estimates and structural-break tests, with the multiple-comparison problem acknowledged.
- **Establish the benchmark —** a random walk and a seasonal naive, reported alongside every later model for the rest of the project.
- **Add structure —** add exogenous drivers only where step 3 produced a stable, interpretable relationship.
- **Backtest —** rolling-origin backtest at four horizons, scored against the benchmark with the metrics in Section 5.5.
- **Interpret —** translate the numbers into a decision, a range and a stated condition under which the conclusion would change.

5.3 The model ladder

Models are added in order and each must beat the one below it on the same backtest. A complex model that ties with a simple one is discarded, because it costs more to explain and more to maintain.

Table 23. *Six modelling levels. Each must beat the level below it to be retained.*

Level	Method	What it is for	Minimum data requirement
L0	Naive pair: random walk (last month) and seasonal naive (last year, same month)	The benchmark every result is measured against; both are always reported	One year of monthly history
L1	ETS and STL plus ARIMA	Captures trend and seasonality without exogenous inputs; the robust default	Five years of monthly history
L2	SARIMAX with weather and price drivers	Tests whether the leading indicators add measurable skill	Aligned exogenous series with no gaps at the origin
L3	Gradient boosting (LightGBM) on engineered features	Captures non-linear interactions and many weak drivers	At least five years and fifty observations per series
L4	Inverse-MASE weighted ensemble of L1–L3	Reduces variance where the models fail in different ways	Backtest outputs from the lower levels
L5	Scenario overlay using elasticities	Answers policy and weather what-ifs that no statistical model can identify	A parameter table with a cited source per elasticity

5.4 Feature engineering rules

Feature engineering is where agricultural models most often cheat. The rules below are enforced in code, not by discipline.

- **Availability first —** every feature must be computable at the forecast origin. A variable published after the origin is excluded, however predictive it looks.
- **Lags and rolling windows —** lags at 1, 2, 3, 6 and 12 months; rolling means and standard deviations over 3, 6 and 12 months, always shifted so the window ends before the origin.
- **Weather is stage-specific —** cumulative rainfall, heat-stress days and dry-spell length computed inside the crop's own phenological windows, not by calendar month.
- **Vegetation is an anomaly —** NDVI anomaly at three dekads before the origin, standardised against the same period in the previous five years.
- **Seasonality is explicit —** Fourier terms or month dummies, chosen once and applied consistently across crops so results remain comparable.
- **No leakage —** no target-derived feature, no future value, and no transform fitted on the full history. A unit test asserts that every feature's timestamp precedes the origin.

5.5 Evaluation protocol

Forecasts are evaluated by rolling origin, never by a random split. The first training window covers 36 months; the origin advances one month at a time for twelve steps, and each step forecasts 1, 3, 6 and 12 months ahead. Transforms are refitted inside every fold, which is the only way the reported error resembles the error a user would experience.

Table 24. *Seven evaluation metrics, each answering a different question.*

Metric	What it measures	Target	Why it is included
MASE	Error scaled against the in-sample seasonal naive error	< 1.0 against the better naive benchmark	Scale-free, so crops with different price levels can be compared
sMAPE	Symmetric percentage error	Crop-specific, 10–20% at three months	The number a non-specialist reader can interpret
RMSE	Error in ₹ per quintal	Reported, not targeted	Expressed in the units of the actual decision
Pinball loss	Accuracy of the 10th, 50th and 90th percentiles	Minimised	Judges the interval, not only the point
Interval coverage	Share of actuals falling inside the 80% interval	80% ± 5 points	Catches overconfident forecasts
Directional accuracy	Share of months where the sign of change is correct	≥ 55%	The measure a trader actually acts on
Bias	Mean error as a share of mean price	|bias| < 5%	Catches a model that is consistently optimistic

THE PROMOTION RULE

A model replaces the benchmark only if it beats the better of two naive forecasts — a random walk and a seasonal naive — on MASE, with interval coverage within five points of nominal and directional accuracy of at least 55%, on the same folds, for that crop. The pilot in Appendix F is why the pair is required: measured against seasonal naive alone, a seasonal model cleared every threshold while a random walk still beat it in 45 of 60 crop-horizon cells. If no model clears the bar, the report publishes the naive forecast plus a driver analysis. That is a legitimate result and will be presented as one.

5.6 Uncertainty and explanation

Every published forecast is an interval first and a point second. Intervals are produced from bootstrapped residuals for the statistical models and from quantile loss for the gradient-boosted model, with coverage checked against the backtest rather than assumed. Explanations use coefficients with confidence intervals for the regression models and SHAP values for the tree model, aggregated to the driver level so that a reader sees 'rainfall deficit' rather than forty correlated lags. Feature importance is reported with its instability across folds, because an importance that changes sign between folds is noise dressed as insight.

5.7 Scenario design

Scenarios answer the questions a point forecast cannot. Five are planned, each with an explicit transmission channel and a parameter drawn from a named source rather than invented.

Table 25. *Five scenarios, their channels and the output each produces.*

Scenario	Shock	Transmission channel	Output
Deficient monsoon	Rainfall −20% against long-period average in the reproductive window	Yield → arrivals → price	Price range and margin impact by crop
Export restriction	Crop moved from open to prohibited for one marketing year	Domestic supply → price wedge vs world	Domestic–world price gap; exporter exposure
Fertiliser cost spike	Urea and DAP prices +20%	Cost of cultivation → acreage next season	Margin compression; planting response
Fuel and freight shock	Diesel +15%	Transport wedge → spatial dispersion	Inter-mandi basis widening
Support price increase	MSP +5% with active procurement	Price floor → market price and stocks	Floor effectiveness; fiscal exposure

5.8 Tooling and reproducibility

Table 26. *The stack, chosen for cost, reproducibility and familiarity.*

Layer	Tool	Purpose
Ingestion	Python requests and scheduled scripts	Pull APIs and files; write immutable raw snapshots with checksums
Storage	Parquet files plus DuckDB	Columnar storage and SQL over local files without a server
Transformation	pandas, with polars for large pulls	Cleaning, harmonisation and panel construction
Statistics	statsmodels	STL, ARIMA, SARIMAX, error-correction models
Machine learning	LightGBM and scikit-learn	Gradient boosting, preprocessing pipelines, cross-validation
Explanation	SHAP	Driver attribution for the tree model
Visualisation	matplotlib and plotly	Static report figures and interactive exploration
Delivery	Streamlit	A single-page indicator and forecast dashboard
Version control	Git, with data hashes in a manifest	Code history plus a verifiable link from result to data snapshot
Environment	Python virtual environment with pinned requirements	The same run produces the same numbers on another machine

Reproducibility is treated as a feature, not a courtesy. The repository contains a single entry point that rebuilds the panel from raw snapshots, a manifest recording a SHA-256 hash for every input file, seeds fixed for every stochastic step, and a run log recording the environment. A clean rebuild must complete in under ten minutes on a laptop; if it does not, the pipeline is too heavy for a five-week project.

5.9 Model risk register

Table 27. *Eight model risks and the control that limits each.*

Risk	How it would show up	Control
Overfitting	Excellent in-sample fit, worse than naive out of sample	Rolling-origin evaluation; complexity added only on evidence
Leakage	Implausibly low error that collapses on new data	Availability-date test on every feature; transforms fitted inside folds
Non-stationarity	Coefficients that change sign between sub-periods	Difference or detrend; report subsample stability
Short series	Wide intervals that make the forecast useless	Minimum observation rule; pool across similar states where justified
Structural break	A step change no driver explains	Break tests; scenario mode after a policy event
Spurious drivers	A high correlation with no mechanism	Require an economic mechanism before a driver enters the model
Overconfident intervals	Coverage well below nominal	Conformal calibration on the backtest residuals
Unstable importance	Feature rankings that reshuffle every fold	Report importance with variance; do not narrate single-fold rankings

6. Step-by-Step Execution Plan

The work is organised into nine phases. Each phase has a defined output and a quality gate: the gate is not a formality, because every later phase assumes the previous output is trustworthy. A failed gate is handled by narrowing scope, not by proceeding.

6.1 Phase overview

Table 28. *Nine phases, twenty-four working days.*

Phase	Focus	Output	Days
P0	Scope, source register, repository and conventions	Signed-off plan and empty pipeline skeleton	2
P1	Acquisition of every critical series	Immutable raw snapshots with manifest	4
P2	Profiling and quality assessment	Profiling report and defect log	2
P3	Cleaning and harmonisation	Clean tables, dictionary and cleaning log	4
P4	Indicator computation and feature build	Analysis-ready panel with 30 indicators	3
P5	Exploratory analysis and hypothesis tests	EDA report with six core exhibits	2
P6	Baseline and candidate models with backtests	Model artefacts and backtest table	3
P7	Evaluation, scenarios and interpretation	Evaluation report and scenario grid	2
P8	Reporting, dashboard and handover	Memo, dashboard, reproducibility pack	2

6.2 The pipeline at a glance

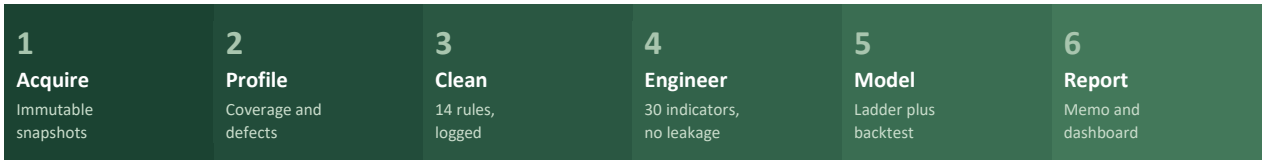


Figure 1. *The six-stage pipeline. Each stage has a gate that must pass before the next begins.*

6.3 Phase detail

P0 — Scope, register and set-up (2 days)

- **Scope —** freeze the crop, state and horizon scope; record any later change in a change log.
- **Source register —** one row per source with publisher, endpoint, licence, refresh rate, owner and known defects.
- **Conventions —** folder structure, naming conventions, unit table, crop master, LGD geography table, and a README that a stranger could follow.

Output: this plan, plus a repository that ingests one sample series end to end. Gate: the sample series reproduces byte-for-byte from a clean clone.

P1 — Acquisition (4 days)

- **Order matters —** four sources first — Agmarknet, DES, IMD and FAOSTAT — because they cover the must-have indicators and expose the hardest formatting problems earliest.
- **Snapshot, don't overwrite —** each pull writes a dated file plus a row in a manifest with URL, timestamp, record count and SHA-256 hash. Raw files are never edited.
- **Respect the source —** throttled and identified requests, with a retry and backoff policy; failures logged rather than silently skipped.
- **Access check —** confirm reachability from an Indian network for the ministry portals that were not verifiable from the audit environment.

Output: raw corpus covering at least 90% of planned series. Gate: every file in the manifest can be re-downloaded and hash-matched.

P2 — Profiling and quality assessment (2 days)

- **Profile every table —** row counts, date ranges, duplicate keys, null patterns, value distributions and cardinality of every categorical field.
- **Defect log —** record each defect with a severity, an example row and the rule that will resolve it. The log is the audit trail for every later decision.
- **Reconcile sources —** cross-source comparison for production and prices; disagreements above 5% are investigated, not averaged.

Output: profiling notebook and a prioritised defect log. Gate: no critical series with an unexplained coverage gap below 90%.

P3 — Cleaning and harmonisation (4 days)

- **Rules, not ad-hoc edits —** fourteen rules in Section 6.4, applied in a fixed order and each implemented as a testable function.
- **Change log —** every dropped, imputed or corrected value is written to a side table with the rule that touched it, so the effect of cleaning is measurable.
- **No silent loss —** row counts before and after each step; a step that removes more than 2% of rows requires a written justification.

Output: clean tables, a data dictionary and a cleaning log. Gate: the validation suite passes, and the panel rebuilds deterministically.

P4 — Indicators and features (3 days)

- **Must-haves first —** the twelve must-have indicators first, each with its formula in code and in the dictionary.
- **Feature factory —** lags, rolling windows, stage-specific weather and vegetation anomalies, all generated by one function so the availability rule is enforced once.
- **Leakage test —** assert that no feature timestamp follows its forecast origin; fail the build if it does.

Output: analysis-ready panel with thirty indicators. Gate: leakage test passes and every indicator has a source and an owner.

P5 — Exploratory analysis (2 days)

- **Six exhibits —** coverage and gaps, price distributions by crop and state, seasonality, correlation structure and the first lead–lag evidence.
- **Hypotheses —** test H1 to H4 with the methods named in Section 2.4 and record negative results.

Output: EDA report. Gate: each exhibit answers a stated question and carries its source and n.

P6 — Modelling and backtesting (3 days)

- **Ladder in order —** benchmark first, then ETS, then the exogenous model, then gradient boosting; each scored on identical folds.
- **Honest validation —** rolling origin, twelve steps, four horizons; no random splits anywhere.
- **Artefacts —** save the fitted objects, the fold-level errors and the seeds, not only the final chart.

Output: model artefacts and a backtest table by crop and horizon. Gate: at least one model beats both naive benchmarks on four of six crops at three months, or the fallback is invoked.

P7 — Evaluation and scenarios (2 days)

- **Where it fails —** error decomposition by crop, state, horizon and season; look for the pattern in the failures, not only the average.
- **Scenario grid —** run the five scenarios and translate each into a margin effect with a stated elasticity.
- **Falsifiers —** state the conditions under which the recommendation reverses.

Output: evaluation report and scenario grid. Gate: every claim traceable to a metric, a table or a cited parameter.

P8 — Reporting and handover (2 days)

- **Decision memo —** two pages: the finding, the range, the decision, the caveat.
- **Dashboard —** one page in Streamlit: current prices, forecast with intervals, indicator heat-map, data freshness.
- **Handover pack —** README, pinned environment, manifest and a ten-minute rebuild; plus a one-page handover note for the next intern.

Output: memo, dashboard and reproducibility pack. Gate: a reader who has never seen the project can rebuild the panel and explain the forecast.

6.4 Cleaning and harmonisation rules

Fourteen rules, applied in order. Each is written as a testable function so that the same input always produces the same output, and so that a reviewer can see exactly what was changed and why.

Table 29. *The fourteen cleaning rules, in application order.*

Issue	Typical example	Rule applied
Duplicate quotes	The same mandi, crop, date and variety appearing twice after a re-run	Keep the latest retrieval from the same source; log superseded rows; never sum duplicates
Unit drift	₹ per kg vs ₹ per quintal; tonnes vs thousand tonnes; acres vs hectares	Canonical SI at ingestion; conversion table held in code (1 quintal = 100 kg; 1 tonne = 10 quintals; 1 hectare = 2.471 acres)
Commodity aliases	'Paddy(Dhan)', 'Rice', 'Wheat FAQ', regional names	Canonical crop master with a synonym list and a separate variety flag
Geography changes	Telangana created in 2014; districts renamed or split	LGD codes as the join key, with a mapping table carrying effective dates
Date conventions	DD/MM/YYYY vs ISO; fiscal year vs calendar year	ISO 8601 plus explicit fiscal_year, crop_year and marketing_year columns
Missing values	Blank, zero and 'not traded' all appearing as empty cells	Classify every gap by cause; no blanket imputation; rule chosen per indicator
Zero or null prices	Market closed, holiday, or no trade that day	Treated as missing, never as zero; imputed only within mandi–crop for gaps of seven days or less, otherwise dropped

Issue	Typical example	Rule applied
Outliers and typos	₹400 recorded instead of ₹4,000	Robust z-score on log price within mandi–crop; flag, review the top twenty manually, and correct only with evidence
Non-trading days	Sundays, festivals, weekly market closures	State-specific trading calendar; forward fill only for plotting, never for modelling
Official revisions	Advance estimate replaced by second and then revised estimate	Every release stored with vintage_id and retrieved_at; models trained on the vintage available at the origin
Nominal vs real	Inflation mistaken for a price signal	CPI deflator applied; both nominal and real series reported
Aggregation	Averaging district yields to get a state yield	Ratio of sums: total production ÷ total area, never a mean of ratios
Currency	USD series joined to INR series without conversion	INR canonical; dated RBI reference rate for any conversion, stored with the row
Suppression	Small counts withheld by the source	Retained and flagged; never inferred or back-calculated

6.5 Data dictionary specification

The dictionary is a deliverable in its own right. The table below defines the columns every fact table carries; the indicator tables add one row per indicator with its formula and unit.

Table 30. *Core fields carried by every fact table.*

Field	Type	Example	Rule
period	date	2026-07-01	First day of the month or the year of record
period_type	enum	month	month, year, week or event
state_lgd	integer	27	LGD state code; never a name
district_lgd	integer	509	LGD district code; null for state-level facts
mandi_id	text	MH_PUNE_01	Register-issued identifier; null for non-market facts
crop_code	text	WHEAT	Crop master key, never a free-text label
variety	text	FAQ	Source variety where reported; null otherwise
indicator_code	text	MODAL_PRICE	One row per indicator in the dictionary
value	float	2410.0	Canonical unit for the indicator; never a formatted string
unit	text	INR/quintal	Controlled vocabulary, checked against indicator master
source_id	text	AGMARKNET	Foreign key to the source register
vintage_id	text	2026-07-14	Publication date of the release the fact came from
retrieved_at	timestamp	2026-10-27T09:14Z	When the snapshot was taken
quality_flag	enum	ok, imputed, outlier, provisional	Never null; drives what is excluded from a model

6.6 Interpretation protocol

Cleaning produces data; interpretation produces a claim. The protocol below is what separates the two, and it is applied to every exhibit before it reaches the memo.

- **State the question first —** the question is written above the chart, not inferred from it.
- **Compare, don't exclaim —** a 6% rise means nothing until it is compared with the seasonal norm, the five-year average and the benchmark forecast.
- **Interval before point —** report the interval, then the point, then the caveat.
- **Know which quantity —** a level, a growth rate and a share are three different claims; never substitute one for another in a sentence.
- **Stress the result —** rerun the headline result without the top 5% of markets by volume. If it disappears, it was a market artefact.

- **Name the mechanism —** a coefficient is a measurement, not an explanation; name the mechanism or label the finding as unexplained.
- **Separate three registers —** what the data shows, what it suggests, and what it cannot say, in that order.

7. Timeline and Milestones

The plan is scheduled against the internship calendar rather than an abstract sprint. Five calendar weeks run from 22 October to 26 November 2026, and the five weekly tasks are treated as the deliverables of the five weekly gates. That alignment is deliberate: each week's submission is the next phase of this plan, not a parallel essay.

7.1 Planning assumptions

- **Capacity —** approximately 3.5 focused hours on each working day, giving 24 scheduled days and roughly 84 hours in total.
- **Calendar —** 26 weekdays fall inside the internship window; one is reserved for the Diwali holiday period, leaving 25 usable days against 24 scheduled.
- **Dependencies —** public data only, no procurement, no waiting on third-party approvals.
- **Team —** one analyst performing every role; review happens at the weekly gate rather than through a separate QA function.
- **Scope —** the scope in Section 1.1 is fixed after M0. Any addition goes into a change log and is traded against something already planned.

7.2 Workstream schedule

Shaded cells show where each workstream is active. Milestone markers sit in the week the gate is reviewed.

Workstream	W1 22–28 Oct	W2 29 Oct–4 Nov	W3 5–11 Nov	W4 12–18 Nov	W5 19–26 Nov
P0 Scope and set-up	◆ M0				
P1 Acquisition	■	◆ M1			
P2 Profiling and QA		■			
P3 Cleaning		■	◆ M2		
P4 Indicators and features			■	■	
P5 Exploratory analysis			■		
P6 Modelling and backtest				◆ M3	
P7 Evaluation and scenarios				■	■
P8 Reporting and handover					◆ M4

■ active workstream ◆ milestone gate

Figure 2. *Workstream schedule, 22 October to 26 November 2026. Grey cells are outside the active window.*

7.3 Milestones and acceptance criteria

Table 31. *Five milestones, each with a criterion that can be checked rather than felt.*

ID	Milestone	Date	Acceptance criterion
M0	Scope frozen and plan approved	Wed 28 Oct	Source register has at least 24 entries; scope table signed; repository ingests one series end to end
M1	Raw corpus complete	Wed 4 Nov	At least 90% of planned series snapshotted; every file hash-verified; access confirmed for ministry portals
M2	Analysis-ready panel	Wed 11 Nov	Validation suite passes; 30 indicators computed; leakage test green; dictionary published
M3	Models validated	Wed 18 Nov	Rolling-origin backtest complete for six crops at four horizons; benchmark comparison table produced
M4	Final submission	Thu 26 Nov	Memo, dashboard and reproducibility pack delivered; clean rebuild under ten minutes

7.4 Critical path and dependencies

The critical path runs P1 acquisition → P3 cleaning → P4 features → P6 modelling → P7 evaluation → P8 reporting. Any delay to acquisition moves every later gate, which is why it receives the first four days and why the four hardest sources are pulled first. Exploratory analysis and documentation sit off the critical path and can absorb a lost day without moving a milestone.

- **P1 → P2 → P3 —** the panel cannot be built until cleaning rules are fixed, and the rules cannot be fixed until profiling exposes the defects.
- **P3 → P4 → P6 —** the feature build is blocked by the cleaning gate and blocks every model.
- **P7 → P8 —** the memo cannot be written before the evaluation, but its skeleton is drafted in Week 4 so that Week 5 is editing rather than authoring.

7.5 Weekly cadence

- **Monday —** confirm the week's gate, list the three tasks that would make it fail, and check the data pulls are scheduled.
- **Wednesday —** inspect what arrived, reconcile counts against the manifest, and raise any source that has changed shape.
- **Friday —** run the gate test, update the change log and the risk register, and submit the week's deliverable.
- **Every day —** log hours and decisions, so that the timeline is defended with evidence rather than recollection.

7.6 Contingency and scope reduction

A plan without a degradation path is a wish. If a gate slips, scope is reduced in the order below, and the reduction is recorded in the memo rather than hidden.

Table 32. *Degradation ladder. Scope is cut in this order and never silently.*

Level	Trigger	What is dropped first	What is protected
1	Up to one day lost	Nice-to-have indicators; the interactive dashboard becomes a static page	All must-have indicators and the backtest
2	Two to three days lost	Crops reduced from six to four, dropping cotton and onion	Price and yield forecasts for staples and oilseed

Level	Trigger	What is dropped first	What is protected
3	Four or more days lost	Gradient boosting and scenarios dropped; ETS and benchmark retained	A benchmarked forecast with honest intervals
4	A critical source is unavailable	Substitute recorded in the gap table; the limitation is stated in the memo	Traceability and disclosure

8. Deliverables and Documentation

Ten artefacts are produced. Each has an owner-verifiable acceptance test, because 'a report was written' is not a deliverable specification.

Table 33. *The ten deliverables and how each is judged complete.*

ID	Deliverable	Format	Acceptance criterion
D1	Analysis plan (this document)	DOCX and PDF	Approved at M0; scope and timeline fixed
D2	Source register	XLSX plus CSV	38 sources with publisher, endpoint, licence, refresh rate and defect notes
D3	Raw data corpus and manifest	Parquet and JSON	Every file hash-verified and re-downloadable
D4	Cleaning log and data dictionary	Markdown and XLSX	Every rule applied with before/after counts
D5	Analysis-ready panel	Parquet	30 indicators, leakage test green, deterministic rebuild
D6	Exploratory analysis report	Notebook and PDF	Six exhibits, each with a question, source and sample size
D7	Model artefacts and backtest table	Notebook and CSV	Fold-level errors retained, not only summary scores
D8	Evaluation report and scenario grid	PDF	Every claim traceable to a metric or a cited parameter
D9	Indicator and forecast dashboard	Streamlit	Four panels: current price, forecast with interval, indicator heat-map, data freshness
D10	Decision memo and handover pack	PDF and README	Two pages; clean rebuild under ten minutes by a new reader

8.1 Documentation standards

- **Provenance on the page —** every table states its source, its period and its row count in the caption or the line beneath it.
- **One set of numbers —** numbers in the memo match the dashboard and the notebook to the digit; a single rounding convention is set once.
- **Charts argue —** charts carry a title that states the finding, an axis in real units, and a note on what would make the pattern disappear.
- **A stranger can run it —** the README explains how to rebuild the panel, where the raw snapshots live, and which version of the data produced the published numbers.
- **Limitations are explicit —** limitations are a numbered section, not a footnote, and each one names the direction of the bias it could introduce.

9. Risks and Mitigations

The register below covers project risk rather than model risk, which is handled separately in Section 5.9. Likelihood and impact are assessed qualitatively and reviewed at each gate.

Table 34. *Project risk register. L and I denote likelihood and impact: H high, M medium, L low.*

Risk	L	I	Mitigation	Early warning
A source changes shape or blocks access	M	H	Immutable snapshots; a schema check on every pull; at least two sources per critical series	Record count or column set differs from the previous pull
Rate limits or quota exhaustion	M	M	API keys, throttled requests, nightly batches, cached responses	Repeated 429 responses in the run log
Official revisions invalidate a result	H	M	Vintage store; re-run at each release; intervals widened where revisions are large	Revision above 5% between successive releases
No model beats the naive benchmarks	M	H	Fall back to the random walk with a driver analysis; report the negative result	MASE above the better naive benchmark across all candidate models at M3
Hours lost to coursework or illness	M	H	One day of float; degradation ladder in Section 7.6	Two consecutive days without a completed task
Scope creep toward more crops or states	H	M	Scope frozen at M0; change log; every addition traded against something planned	A new crop appears in a notebook after Week 2
Thin or zero-arrival markets distort the panel	H	M	Minimum-observation rule; arrivals weighting; sensitivity check without the top 5% of markets	More than 20% of market-days with zero arrivals
Terms-of-use or licensing breach	L	H	Public endpoints only; no login scraping; attribution in every output	A source's terms page changes
Reproducibility drift	M	M	Pinned environment, fixed seeds, data hashes, a clean-rebuild test at each gate	Rebuild produces different row counts or totals

10. Governance, Ethics and Limitations

Three obligations run through the project: use public data lawfully, represent it honestly, and leave a trail that lets someone else check the work.

10.1 Data governance

- **One register —** the source register is the single authority on where a number came from; nothing enters the panel without an entry in it.
- **Immutability —** raw snapshots are immutable and dated. Corrections happen downstream, never by editing the original file.
- **No personal data —** no personal or household identifiers are stored; survey microdata is aggregated before it enters the panel.
- **Attribution —** every output credits the originating agency and states the retrieval date.
- **No advice —** the analysis is descriptive and decision-supporting. It is not investment, trading or agronomic advice, and the memo says so.

10.2 Analytical integrity

- **Negative results are results —** results that fail to beat the benchmark are published as failures, with the reason, rather than omitted.
- **Uncertainty is mandatory —** intervals are reported wherever a point estimate appears; a single number without a range implies a precision the data does not support.
- **Three registers —** the memo distinguishes what the data shows, what it suggests and what it cannot say, in that order and under those headings.
- **Traceability —** each published figure carries the identifier of the query and the data version that produced it, so a reader can reproduce it without asking.

10.3 What this analysis will not do

- **Predict policy —** it will not forecast a policy decision, only its plausible effect if taken.
- **Represent all trade —** regulated mandi prices are the observed population; direct, contract and farm-gate transactions are outside it.
- **Promise a year ahead —** twelve-month price forecasts are scenario ranges, not commitments.
- **Speak for an individual —** a forecast of the average price says nothing about the price available to a particular farmer on a particular day in a particular mandi.

11. Resources and Budget

The plan is deliberately cash-light: every source is public and every tool is open source. The binding constraint is time, not money.

Table 35. *Resource requirements. Cash cost is zero unless the optional items are taken.*

Item	Cost	Note
Data sources	₹0	All 38 sources are free at the point of use; no paid feed is on the critical path
Software stack	₹0	Python, DuckDB, statsmodels, LightGBM, Streamlit, Git
Compute	₹0	A laptop is sufficient; the panel is under a few gigabytes
Optional scheduled pulls	₹0 – ₹1,200 / month	A small cloud instance only if pulls must run while the laptop is off
Optional domain for the dashboard	₹900 / year	Cosmetic; not required for the deliverable
Analyst time	84 hours	24 working days at approximately 3.5 hours per day

The time budget is the real budget. At 84 hours, roughly 40% goes to acquisition and cleaning, 25% to features and exploration, 20% to modelling and evaluation, and 15% to reporting. That distribution is intentional: in practice, data work consumes the majority of an agricultural analytics project, and a plan that allocates most of its time to modelling is a plan that will finish late.

12. Project Measures and Conclusion

12.1 How the project itself will be measured

Table 36. *Seven measures for the project, distinct from the model's own metrics.*

Measure	Target	Checked at
Planned series ingested	≥ 90% of the register	M1 and M4
Core panel completeness	≥ 95% after documented rules	M2
Indicators in production quality	30, of which 12 must-have	M2
Crops beating the better naive benchmark at 3 months	≥ 4 of 6	M3
Interval coverage	within 5 points of nominal	M3
Clean rebuild time	under 10 minutes	M4
Traceability	100% of published figures reproducible	M4

12.2 Conclusion

This plan converts a broad brief into a sequence of checkable steps. It fixes six crops, eight states and a twelve-month horizon; it catalogues thirty-eight public sources, twelve of which carry the analysis; it defines thirty indicators and separates them into forecastable, partly forecastable and event-driven; and it commits to a modelling ladder in which both naive forecasts must be beaten before anything more elaborate is reported.

The design is shaped by two judgements about agribusiness data specifically. First, the expensive part is not modelling but reconciliation — the same quantity exists in four places with four different values, and the plan budgets accordingly. Second, forecastability is not uniform: prices can be forecast with intervals, yields only as ranges before harvest, and policy only as scenarios. A plan that blurs those categories will produce confident forecasts that fail in the season they matter most.

Success at the end of Week 5 is therefore modest and verifiable: a panel that rebuilds in ten minutes, a forecast that beats the better naive benchmark on at least four crops, intervals that mean what they say, and a two-page memo that a procurement manager could act on. The four remaining weekly tasks attach directly to this spine — cleaning in Week 2, exploratory design in Week 3, the predictive framework in Week 4 and evaluation in Week 5 — so the plan is also the schedule for the rest of the internship.

Appendix A. Glossary

Terms used in this document that a reader from outside agricultural economics may not share. Definitions are the ones used in the analysis, not the widest ones available.

Table 37. *Terms used throughout the plan.*

Term	Meaning in this plan	Why it matters here
Mandi / APMC	A regulated wholesale market where agricultural produce is auctioned under state marketing rules	Defines the price population; non-regulated trade is outside it
Modal price	The price at which the largest quantity traded in a mandi on a day	The forecast target; more robust than a simple average
Arrivals	Quantity of a commodity reaching a mandi on a given day	The supply signal that leads price at short horizons
MSP	Minimum support price announced by the central government for a marketing year	Acts as a floor and changes the planting decision
Kharif / rabi	Monsoon-sown and winter-sown cropping seasons	Determines which months belong to a crop's marketing window
Gross cropped area	Sum of area sown across all crops in a year, counting multiple cropping	Denominator for input-intensity indicators
A2+FL	Paid-out cost plus imputed family labour, the cost concept used in Indian cost-of-cultivation studies	Base for the cost-per-quintal and margin calculation
Stage of groundwater extraction	Annual groundwater extraction as a percentage of annual recharge, by assessment unit	A resource-risk indicator, not a weather variable
NDVI	Normalised difference vegetation index, a satellite measure of green biomass	Nowcasts crop condition before official yield estimates appear
Stock-to-use ratio	Ending stocks divided by total annual use	The standard predictor of price pressure in grain markets
Basis	The difference between a local cash price and a reference futures or world price	Links domestic prices to global and forward markets
Vintage	A specific published release of an estimate, retained rather than overwritten	Prevents look-ahead bias when backtesting
Nowcast	An estimate of the current state produced before official statistics exist	Fills the publication lag in yield data
MASE	Mean absolute scaled error, where the scaling factor is the seasonal naive error	The benchmark metric, comparable across crops
LGD code	Local Government Directory code for a state or district	The join key that survives renames and splits

Appendix B. Access Notes by Source

Practical notes gathered while checking access. They are the things that cost an afternoon if discovered late, recorded here so that Week 2 does not rediscover them.

Table 38. *Access notes that inform the ingestion design.*

Source	Access path	Practical note
Agmarknet	Public portal; daily price and arrival reports by state, district and market	No documented API on the portal. Prefer the data.gov.in resource for automation; use the portal for spot checks and for varieties the API omits.
data.gov.in	JSON API with a free key; catalogue identifies each dataset by resource id	Default page size is small, so paginate explicitly. Field names and availability change between resources; validate the schema on every pull.
USDA NASS Quick Stats	REST API with a free key; bulk CSV downloads also available	Query by commodity, statistic, year and geographic level. Units differ by commodity, so the unit field must be read, not assumed.
FAOSTAT	Bulk CSV downloads and a documented API	Country and year are the natural keys. Element and item codes are more stable than their labels, which are revised.
World Bank	Open data API and downloadable Excel for the Pink Sheet	The Pink Sheet is revised monthly and re-based periodically; keep the download date with the file.
IMD	Public portals for rainfall, temperature and agromet advisories	Some products are published as PDF or image maps. District rainfall is the practical unit; subdivision data is more complete historically.
NASA POWER	Free API, no key required; JSON and CSV output	Grid resolution is about 0.5°, so a district maps to one or more cells. Record the cell used for each district rather than assuming a match.
Copernicus Data Space	Free registration, then catalogue search and download	Optical imagery is cloud-affected during the monsoon. Plan for gaps and keep a microwave or model-based fallback in mind.
UN Comtrade	Public API with a free quota	HS codes are revised between nomenclatures; map to a single version before building a long series.
eNAM	Public portal with trade and price dashboards	Useful as an independent check on Agmarknet, not as the primary series; coverage of mandis differs.
Ministry portals (DES, CACP, Ag Census)	Free downloads and reports	Reachability varies by network. Confirm access from an Indian connection during Week 2 and cache the files immediately.
CME and OECD	Public websites	Both returned 403 to automated requests during the access check while loading normally in a browser; treat as manual-reference sources.

Appendix C. References and Source Register

All URLs below were checked on 27 October 2026. They are the principal public entry points used in the source register; dataset-level identifiers, access dates and licence notes for all thirty-eight sources are recorded in deliverable D2. Source 6, the World Bank Pink Sheet, was exercised end to end in Appendix F; the others were checked for reachability and access path only.

1. U.S. Department of Agriculture, National Agricultural Statistics Service. *Quick Stats API.* https://quickstats.nass.usda.gov/api
2. U.S. Department of Agriculture, Economic Research Service. *Data products.* https://www.ers.usda.gov/data-products/
3. Food and Agriculture Organization of the United Nations. *FAOSTAT database.* https://www.fao.org/faostat/en/
4. Food and Agriculture Organization of the United Nations. *FAO Food Price Index.* https://www.fao.org/worldfoodsituation/foodpricesindex/en/
5. World Bank. *Agriculture and rural development data.* https://data.worldbank.org/topic/agriculture-and-rural-development
6. World Bank. *Commodity markets: Pink Sheet.* https://www.worldbank.org/en/research/commodity-markets
7. Directorate of Marketing and Inspection, Government of India. *Agmarknet price and arrival portal.* https://agmarknet.gov.in/
8. Government of India. *Open Government Data Platform India.* https://data.gov.in/
9. Ministry of Agriculture and Farmers Welfare. *Electronic National Agriculture Market (eNAM).* https://enam.gov.in/web/
10. Directorate of Economics and Statistics, Government of India. *Agricultural statistics portal.* https://desagri.gov.in/
11. India Meteorological Department. *Mausam portal.* https://mausam.imd.gov.in/
12. India Meteorological Department. *Agromet Advisory Services.* https://agromet.imd.gov.in/
13. National Aeronautics and Space Administration. *POWER project data access.* https://power.larc.nasa.gov/
14. European Space Agency. *Copernicus Data Space Ecosystem.* https://dataspace.copernicus.eu/
15. United Nations Statistics Division. *UN Comtrade database.* https://comtradeplus.un.org/
16. International Trade Centre. *Trade Map.* https://www.trademap.org/
17. European Commission. *Eurostat agriculture statistics.* https://ec.europa.eu/eurostat/web/agriculture
18. National Commodity and Derivatives Exchange. *Commodity prices and contracts.* https://www.ncdex.com/
19. Central Ground Water Board, Government of India. *Groundwater assessment reports.* https://cgwb.gov.in/
20. Pradhan Mantri Fasal Bima Yojana. *Crop insurance dashboards.* https://pmfby.gov.in/
21. Agricultural and Processed Food Products Export Development Authority. *Export statistics.* https://apeda.gov.in/
22. OECD and FAO. *OECD-FAO Agricultural Outlook publications.* https://www.oecd.org/en/publications/
23. CME Group. *Agricultural futures markets.* https://www.cmegroup.com/markets/agricultural.html

Appendix D. Assumptions Log

Every plan rests on assumptions. Recording them with a test and a consequence makes them auditable, and turns a surprise into a documented event rather than an excuse.

Table 39. *Assumptions, their tests, and the consequence of being wrong.*

ID	Assumption	Basis	How it is tested	If it is wrong
A1	Agmarknet modal prices represent the direction of wholesale prices, even if not the level	Widely used in Indian market analysis	Compare with eNAM and retail series for overlapping markets	Report price direction only; withdraw level claims
A2	Eight states are sufficient to characterise national crop behaviour	Production share and data availability	Compare state-weighted aggregate with national series	Add states or restate as an eight-state analysis
A3	District-level yield can be nowcast from weather and vegetation	Published yield-forecast literature	Backtest nowcast against official estimates for three years	Drop the nowcast; rely on trend and banded estimates
A4	CPI is an adequate deflator for agricultural prices	Standard practice in real-price analysis	Compare with a wholesale price index deflator	Report both deflators; note divergence
A5	Mandi coverage bias is stable over time	No evidence either way at planning stage	Track the share of arrivals covered by reporting mandis	Restrict the analysis window to the period of stable coverage
A6	Policy events can be captured as dated flags	Announcements are public and dated	Check flag dates against notifications	Treat the period as a structural break instead
A7	A 3.5-hour daily commitment is available	Stated internship capacity	Weekly hour log against plan	Invoke the degradation ladder in Section 7.6
A8	Free-tier quotas are sufficient for the pull volume	Small number of series, cached locally	Monitor response codes and quota headers	Stagger pulls across days; prioritise the must-have set

Appendix E. Weekly Execution Checklist

A one-page checklist for each week. Items are ticked only when the evidence exists in the repository, not when the task feels finished.

Table 40. *The five weekly gates as a tick-list.*

Week	Checks before the gate can pass
Week 1 **22–28 Oct**	☐ Scope table signed and dated ☐ At least 24 sources registered with licence and endpoint ☐ Repository created with folder conventions and README ☑ One series ingested end to end as a proof of concept (Appendix F) ☐ Unit, crop and geography master tables drafted ☐ This plan submitted
Week 2 **29 Oct–4 Nov**	☐ Four hardest sources pulled and snapshotted ☐ Manifest complete with hashes and row counts ☐ Access confirmed for ministry portals from an Indian network ☐ Profiling notebook run on every table ☐ Defect log prioritised with a rule per defect ☐ Fourteen cleaning rules implemented as functions ☐ Cleaning log and data dictionary published
Week 3 **5–11 Nov**	☐ Validation suite green on the clean panel ☐ Twelve must-have indicators computed and checked ☐ Leakage test passing in the build ☐ Six EDA exhibits complete with source and sample size ☐ Hypotheses H1–H4 tested and results recorded ☐ Stationarity and structural-break checks done
Week 4 **12–18 Nov**	☐ Random walk and seasonal naive scored on all crops ☐ ETS and exogenous models scored on identical folds ☐ Gradient boosting run with nested feature checks ☐ Backtest table complete by crop and horizon ☐ MASE, sMAPE, coverage and directional accuracy reported ☐ Memo skeleton drafted
Week 5 **19–26 Nov**	☐ Scenario grid complete with cited elasticities ☐ Error decomposition by crop, state and season ☐ Dashboard live with four panels ☐ Decision memo limited to two pages ☐ Clean rebuild tested under ten minutes ☐ Handover note written for the next analyst ☐ Limitations section numbered and specific

Appendix F. A Pilot Run on Real Price Data

The four remaining weeks assume that a monthly price series can be assembled, cleaned, decomposed and backtested, and that the resulting errors will be informative. That assumption is cheap to test before any Indian mandi data is touched. This appendix reports the test: the pipeline described in Sections 5 and 6, applied end to end to a public series that was machine-readable on the day this plan was written.

The purpose is not to forecast world prices, which is not the brief. It is to establish three things that the rest of the plan currently only asserts: that the arithmetic produces interpretable numbers, that the benchmark chosen in Section 5.5 can actually be beaten, and that its evaluation rule survives contact with data. The third finding did not survive, and the rule has been amended in Section 5.5 rather than quietly left in place.

F.1 The dataset

The series are the World Bank Commodity Price Data — the Pink Sheet — downloaded on 2 October 2026. The file is monthly, quoted in nominal US dollars, and contained 93 complete monthly observations from Jan 2019 to Sep 2026 at the time of download. Five of the six crops in scope have a global benchmark series: wheat, maize, rice, soybean and cotton. Onion has none, because it is traded almost entirely within national markets; that gap is a limitation of the pilot, not of the plan, and it is the one crop for which only the Indian data will speak.

The file was saved before reading and its download date recorded with it; no value in this appendix is typed by hand, and every number below is computed from that one file.

Table 41. *Five global benchmarks, January 2019 to September 2026. CV is the coefficient of variation; volatility is the annualised standard deviation of monthly log changes.*

Series	Unit	n	Mean	Minimum	Maximum	CV %	Vol. %
Wheat, US SRW	US$/t	93	259.7	197.3 (Apr 2019)	446.7 (Mar 2022)	23.5	26.4
Maize, US	US$/t	93	222.1	143.9 (May 2020)	348.2 (Apr 2022)	24.1	21.7
Rice, Thai 5%	US$/t	93	476.0	356.0 (Oct 2025)	660.0 (Jan 2024)	15.6	19.1
Soybeans	US$/t	93	498.4	340.0 (May 2019)	737.0 (Jun 2022)	21.9	17.5
Cotton, A index	US$/kg	93	2.0	1.4 (Apr 2020)	3.6 (May 2022)	22.4	17.6

The range is the first result, and it is larger than the plan's prose implies. Wheat traded at US$197/t in April 2019 and US$447/t in March 2022, a 2.3-fold move; cotton ran from US$1.40/kg in April 2020 to US$3.60/kg in May 2022. Every series has a coefficient of variation between 15% and 24%, and four of the five exceed 21%. A variable that moves by a fifth of its own mean in a normal year is not one to forecast as a point.

The second result is that the window ends far from where it began. At the last observation, September 2026, every series sits between 27% and 42% below its 2022–24 peak, so a model fitted across the whole window starts from a level the market left two years ago — which is the case for storing vintages and refitting at the origin, not once, that Section 5.4 already makes.

F.2 Worked example 1 — building the seasonal index by hand

The seasonal index is the simplest piece of arithmetic in Section 6 and the easiest to get wrong, so it is done once here in full. For each crop and month, the index is the mean price in that calendar month divided by the mean price over the whole window, multiplied by 100. A value of 100 means the month is average; 107 means it runs seven per cent above the average.

For wheat, December prices averaged US$277.19/t across the 93 months, and the series as a whole averaged US$259.70/t. The December index is therefore 277.19 ÷ 259.70 × 100 = 106.7. The August average is US$236.78/t, giving 91.2. The distance between the two, 15.5 index points, is the seasonal amplitude — the share of the annual price range that repeats on a calendar rather than occurring once.

Table 42. *Seasonal index by calendar month, mean of 2019–2026 = 100. Read across for a crop's year; read down for a month's crops.*

Month	Wheat, US SRW	Maize, US	Rice, Thai 5%	Soybeans	Cotton, A index
Jan	104.7	99.6	102.3	99.3	99.8
Feb	101.4	101.0	99.9	101.3	101.8
Mar	102.8	101.4	98.3	101.3	101.0
Apr	101.0	103.4	100.2	100.4	102.7
May	101.8	104.2	102.0	101.4	103.3
Jun	96.7	102.7	102.6	101.1	101.0
Jul	92.7	99.5	97.8	100.8	98.3
Aug	91.2	92.7	100.8	97.2	98.5
Sep	96.7	96.7	100.5	99.0	98.7
Oct	100.3	100.1	95.7	97.2	97.7
Nov	105.7	99.1	96.8	100.6	98.6
Dec	106.7	99.5	102.5	100.0	97.8

Only wheat has a clean harvest shape: a July–September low after the northern-hemisphere harvest and a December–January high, which is what a storage model predicts. Maize peaks in May and troughs in August; cotton peaks in May and troughs in October. Rice is nearly flat, with an amplitude of 6.9 points, and that flatness is informative rather than reassuring: Thai 5% is an export price set by policy and by the availability of Indian and Vietnamese supply, not by a harvest calendar. Its two largest deviations in the window — a January 2024 peak of US$660/t and an October 2025 trough of US$356/t, a 46% fall — were not seasonal events at all.

That distinction has a direct consequence for the model ladder. A seasonal term fitted to wheat or maize is capturing something that repeats and will help. Fitted to rice or onion it absorbs a policy shock, labels it seasonal, and forecasts the shock repeating in the same month next year. Seasonality is therefore fitted per crop rather than once for the panel, and its strength is reported alongside the forecast.

F.3 Worked example 2 — what the seasonal carry is actually worth

A seasonal index is only useful if the movement it describes is large enough to act on. The test is a carry calculation: buy at the seasonal trough, hold, sell at the peak, and compare the gross spread with the cost of holding. For wheat, buying at the August mean of US$236.78/t and selling at the December mean of US$277.19/t yields US$40.41/t, or 17.1% over four months. That is 4.3% a month, or about US$10.10/t a month of storage.

Table 43. *Trough-to-peak carry from the monthly means, and the same spread expressed as a rate per month held.*

Series	Peak	Trough	Carry	Gross spread	Held, months	Per month
Wheat, US SRW	Dec	Aug	17.1%	US$40.41/t	4	4.3%
Maize, US	May	Aug	12.4%	US$25.55/t	9	1.4%
Rice, Thai 5%	Jun	Oct	7.3%	US$33.19/t	8	0.9%
Soybeans	May	Aug	4.3%	US$20.75/t	9	0.5%
Cotton, A index	May	Oct	5.6%	US$0.11/kg	7	0.8%

The comparison is what matters. Wheat offers US$10.10/t a month; rice offers US$4.15/t a month over eight months, and soybean US$2.31/t a month over nine. Storage, insurance, shrinkage and the cost of money must all fit inside those numbers. For wheat they might; for soybean they almost certainly do not, which means the seasonal pattern there is a description of the market rather than a trading opportunity, and should not be presented as one.

The same arithmetic gives the Week 3 test a concrete form: compare the observed trough-to-peak spread in a district with the warehousing rate actually quoted to farmers there. If the spread is narrower than the rate, the finding is not that the data are noisy but that the trade does not exist in that district — worth more to a procurement manager than another percentage point of forecast accuracy.

One caution belongs here. This index is a single average across 2019–2026, and 2022 dominates the variance; an index computed from 2022 onward would show a different shape. That is why the backtest refits the seasonal component at every origin, and why a published index always carries its window and its vintage.

F.4 Worked example 3 — the benchmark test, and the rule that failed it

The evaluation protocol in Section 5.5 was applied unchanged. Training begins with a 36-month window, the origin advances one month at a time, and at every origin three models forecast one to twelve months ahead: a random walk that repeats the last observed price, the seasonal naive forecast that repeats the same month last year, and an exponential smoothing model with an additive seasonal term and no trend, with its parameters estimated inside each fold. Errors are scored as MASE, with the scaling factor computed inside each fold from that fold's own training window. With 45 origins, twelve horizons and five crops, each model produced 2,700 forecasts, and the comparison covers 60 crop-horizon cells.

Table 44. *Out-of-sample MASE by horizon. Below 1.0 means the model beats the in-sample seasonal naive error; lower is better.*

Model and horizon	Wheat, US SRW	Maize, US	Rice, Thai 5%	Soybeans	Cotton, A index
1 month — Random walk	0.28	0.21	0.27	0.20	0.14
1 month — Seasonal naive	1.17	0.91	1.37	0.95	0.93
1 month — ETS, additive season	0.33	0.21	0.25	0.23	0.19
3 months — Random walk	0.58	0.36	0.49	0.33	0.33
3 months — Seasonal naive	1.09	0.86	1.38	0.91	0.85
3 months — ETS, additive season	0.59	0.39	0.52	0.39	0.43
6 months — Random walk	0.56	0.47	0.76	0.47	0.40
6 months — Seasonal naive	0.95	0.81	1.38	0.87	0.69
6 months — ETS, additive season	0.56	0.51	0.80	0.53	0.50
12 months — Random walk	0.85	0.71	1.43	0.84	0.63
12 months — Seasonal naive	0.85	0.71	1.43	0.84	0.63
12 months — ETS, additive season	0.85	0.71	1.43	0.84	0.63

Against the benchmark this plan originally specified, the seasonal model is decisive. It beats seasonal naive on 90.8% of origins at one month, on 77.8% at three months and on 72.4% at six, and it cuts MASE by 72–82% at one month and by 28–42% at six. On those numbers it would pass the promotion rule comfortably.

Against a random walk it does not. Across the sixty crop-horizon cells the random walk was better in 45, tied in 10 and worse in 5. At one month the only crop where the seasonal model wins is rice, and by 0.02 MASE — inside any reasonable noise band. A model that beats the stated benchmark by three-quarters and still loses to last month's price is not a good forecast; it is a badly chosen benchmark.

Table 45. *Share of origins on which the seasonal ETS forecast had a smaller absolute error than seasonal naive, in per cent.*

Series	1 month	2 months	3 months	6 months	9 months	12 months
Wheat, US SRW	87	76	69	62	62	40
Maize, US	89	80	76	71	58	44
Rice, Thai 5%	96	93	91	89	69	31
Soybeans	91	87	82	80	76	49
Cotton, A index	91	80	71	60	58	47

The third result is arithmetic rather than empirical. At twelve months the three models produce the same forecast: the largest MASE difference between any two of them is 0.000. This is not a coincidence. The seasonal naive forecast for month t+12 is the value observed at month t, which is exactly the random walk's forecast at that horizon, and an additive seasonal ETS with no trend converges to the same point. At twelve months, MASE cannot separate models because there is nothing distinguishing them to separate. The win rate at that horizon falls to 42.2%, which is a coin toss, as it must be.

This is the quantitative justification for a promise the plan already made in Section 1.1 on judgement alone: twelve months is a scenario range, not a point forecast. A report that claims skill at that horizon on this evidence would be measuring its own arithmetic.

F.5 What the pilot changes

Three amendments follow, and all three are carried into Sections 5 and 12 rather than left here as an anecdote.

- **The benchmark is a pair, not a single series —** A model must now beat the better of two naive forecasts — a random walk and seasonal naive — not seasonal naive alone. The gate in Section 5.5 is rewritten accordingly, and the margins remain MASE below 1.0, interval coverage within five points of nominal, and directional accuracy of at least 55%.
- **Short-horizon skill must be measured against persistence —** Seasonal naive is a weak test at short horizons, because repeating a price from twelve months ago is a large move when prices have trended. Against a random walk the honest question is whether seasonality adds anything beyond the current level. On global benchmarks it added little, even where the amplitude in Table 43 is real, because year-to-year level shifts are larger than the seasonal swing. On Indian mandi data the harvest glut and the MSP date may change that, and Week 4 tests it.
- **A negative result is a result —** The plan now states in advance what happens if the Indian panel behaves as the global one did: the deliverable is a random-walk-plus-drivers forecast with a driver analysis, published as the result. A negative result about seasonality, demonstrated on 2,700 forecasts, is worth more than a seasonal model that cannot be defended.

> **WHAT THE PILOT ESTABLISHES**
>
> The arithmetic in Section 6 works end to end on a raw public file: the seasonal index, the carry calculation and the rolling-origin backtest all produce numbers a manager could use. It also establishes where the original evaluation rule was too generous, which is the more useful of the two findings. It does not establish anything about Indian prices, and it is not offered as evidence about them.

F.6 What the pilot cannot tell you

Five limits bound the result, and each one is a Week 2 or Week 3 task rather than an excuse.

- **Different price population —** Global benchmark prices are not mandi prices. They are FOB quotations for traded grades, set by export supply and by freight, and they relate to a district modal price through a

basis that varies by state, season and policy. The plan's claim that global series are a sanity check and not training rows is unaffected by this exercise.

- **Nominal, not real —** The series are nominal US dollars. A rise in nominal dollars can be a fall in real terms, which is why the deflation rule in Section 6.4 is not optional; the pilot sidesteps it by making no claim about real prices.
- **Five of six crops —** Onion has no global series, so the crop where seasonality matters most to an Indian reader is the one the pilot says nothing about.
- **One shock drives the variance —** Ninety-three months contain one dominant shock. The 2022 spike accounts for a large share of the variance in every series, so these errors are an average across a calm period and an extreme one, not a typical year.
- **A narrow model set —** Only two model classes were taken to the backtest. The ladder in Section 5.3 adds ARIMA, exogenous regressors and gradient boosting; the pilot establishes the benchmark, not the ceiling.

F.7 Reproducing this appendix

Nothing in these tables depends on a step that is not written down, and the recipe is short enough to state in full.

- **One file —** download the monthly historical file from the World Bank commodity-markets page and record the date on the saved copy; the vintage used here was 2 October 2026.
- **Five columns —** read the monthly prices sheet, keep the soybean, maize, Thai 5% rice, US SRW wheat and cotton A index columns, parse the YYYY Mmm index into dates, and take every row from January 2019 onward.
- **One adjustment —** compute the monthly means, divide each by the window mean and multiply by 100. The seasonal index table above is that computation and nothing else.
- **Three models, 45 origins —** train on 36 months, advance the origin one month at a time to September 2025, and fit three models at each origin: persistence, seasonal naive, and additive seasonal exponential smoothing with no trend and parameters estimated inside the fold.
- **One score —** MASE, scaled by the fold's own in-sample seasonal naive error, with fold-level errors kept rather than only their average — an average can look respectable while a third of the folds fail.

A reader with the file and those five steps reproduces every number in this appendix to the last digit shown. That is the standard the final report will be held to, and it is the reason the pilot was worth a day of Week 1 rather than a paragraph of assertion.